Lecture 3: Fundamental Probability Concepts for Risk Management and Actuarial Science

Reference Notes

Project Title:
Nurturing Future Risk Managers and Actuaries:
An AI-powered Journey for Gifted Secondary Students

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Contents

Link to Risk Management & Actuarial Science

What this handout is for: These notes are written for students. They are designed to help you move step by step from basic probability language to risk-based thinking, with worked examples, technology support, and short activities for practice.

Structural Core (⋆)

Depth markers: ⋆ Core ideas you must know ⋆⋆ Insight that helps you explain why a rule works ⋆⋆⋆ Extension for students who want an extra challenge.

Pause-and-Think

As you read, keep asking: *How do words like "and", "or", and "given that" change the mathematics?* This habit will help you choose the correct probability rule with confidence.

1 Why this lecture matters

Many real-life situations in finance and insurance can be phrased as questions about uncertainty: Will a claim occur? How likely is a severe event? If a machine alarm goes off, how worried should we really be? Can two events be unable to happen together and still fail to be independent? Can a strategy be *fair on average* but still be a bad idea? In this lecture we begin with words and diagrams, move to rules and calculations, and end with random variables, probability distributions, expectation, and variance — the language actuaries and risk managers use to turn uncertainty into decisions.

Technology will be used throughout. Google Gemini can help you generate examples, explain steps, suggest spreadsheet formulas, and draft R code, while **R** can simulate long-run behaviour and check by-hand calculations. An algebraic approach is possible, but technology often makes patterns easier to see and interpret. However, technology does not replace judgement: if the model is silly, the software will still compute a silly answer very accurately.

Learning outcomes

By the end of this lecture, students should be able to:

1. define experiment, outcome, event, sample space, and simple set notation in risk contexts;
2. distinguish experimental probability from theoretical probability;
3. apply complement, addition, multiplication, and conditional probability rules;
4. explain what independence means, why it matters, and why it must not be guessed by intuition alone;
5. distinguish mutually exclusive events from independent events;
6. use Venn diagrams and tree diagrams as thinking tools rather than decoration;
7. define a discrete random variable and construct a simple probability distribution in table, graph, and formula form;
8. compute and interpret expected value, variance, and standard deviation in risk language;
9. recognise the transformation rules for expectation and variance;
10. describe Bayes' theorem as a natural extension of conditional probability.

Cognitive Trigger

You probably already have an intuition for what "50% chance" means. But could you explain, in strict mathematical language, *why* the probability should be 50%, *what the sample space is*, and *what assumptions you have silently made*? Real risk work begins where vague intuition stops being enough.

Google Gemini Exploration (Copy & Paste Prompt)

Prompt you can copy and adapt:

```
Create three short real-life stories where the words "and", "or", and "given that" lead to
different probability calculations.  For each story, explain which set operation or conditional
probability idea is involved, and include one common student mistake.
```

2 Opening Problem: A fair insurance product, or a dangerous illusion?

Opening Problem

A school is considering a very simple **event disruption insurance** contract for an outdoor showcase night. For each event, the organiser must choose one of three strategies:

1. **Do nothing**: pay all disruption losses out of school funds.
2. **Buy insurance**: pay a premium of HK$8,000 and receive compensation if disruption occurs.
3. **Move indoors**: remove most weather risk, but lose sponsorship and visitor excitement.

Past data from 50 comparable dates show the following outcomes.

Weather outcome	Frequency	School loss if outdoors	Insurance payout	Loss if moved indoors
Dry / calm	28	HK$0	HK$0	HK$16,000
Light rain	12	HK$9,000	HK$5,000	HK$8,000
Heavy rain	7	HK$30,000	HK$24,000	HK$2,000
Cancellation / typhoon	3	HK$92,000	HK$80,000	HK$0

Motivating questions

1. What is the probability of *serious disruption* (heavy rain or cancellation)?
2. If rain has already started, how should that change the organiser's view of cancellation risk?
3. Which strategy has the best *average* financial outcome? Which has the smallest *spread* of possible outcomes?
4. If a strategy is fair in expected value but exposes the school to a tiny chance of a disastrous loss, is it still acceptable?
5. If the insurer says, "Heavy rain and cancellation are both weather problems, so we can treat them as basically independent," what should your first reaction be?

Investigation / Activity: Think first, calculate later

Before using any formula, decide:

- Which questions are about **history**, and which are about a **model**?
- Which are about single events, and which are about long-run averages?

- Which choice might a cautious risk manager prefer? Which might a very ambitious event organiser prefer?
- What crucial information is still missing from this table if you wanted a genuinely robust decision?

Thought Trigger

A gamble can be "fair" on average and still be strategically foolish. Risk management begins exactly where average thinking becomes too shallow.

3 A. Probability language: from uncertainty to structure

Cognitive Trigger

You roll a standard 6-sided die. What is the probability of rolling a number that is both *even* and *prime*? Why is ordinary language dangerous here if the sample space and the exact event definitions are not made explicit?

Probability is not guesswork. It is a numerical language for uncertainty. To use it well, we need a clear structure.

Core definitions

- **Experiment**: a process with uncertain outcome (rolling a die, next day's weather, whether a policyholder makes a claim).
- **Outcome**: one possible result of the experiment.
- **Sample space** S: the full collection of possible outcomes.
- **Event** A: any subset of the sample space that is relevant to the decision-maker.

Risk / actuarial interpretation: The event is chosen by the *decision question*, not by the mathematics alone.

Set language you will use in this lecture

To describe events clearly, we will use set language including **union**, **intersection**, and **complement**. In this lecture, the symbols

$$A \cup B, \qquad A \cap B, \qquad A',$$

not as optional extras but as normal working language for probability.

Structural Core (⋆)

The foundational rule: if all basic outcomes are equally likely, then

$$\mathbb{P}(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S}.$$

In this sense, probability begins as a counting problem before it becomes a modelling problem.

Paired Examples: Works vs Fails

Works: Roll a fair die. Let $A = \{2, 4, 6\}$. Then $P(A) = 3/6$.
Fails if used carelessly: "The chance of traffic delay tomorrow is 50% because half the days I remember felt slow." Memory is not a well-defined sample space.

Worked Example

Problem. A one-year device warranty can end in exactly one of four ways: no claim, screen repair, battery replacement, or full replacement. Translate this risk story into a sample space and two useful events.
Micro-steps

Step 1: Define the experiment: observe the warranty outcome over one year.

Step 2: List the sample space:

$$S = \{\text{No claim}, \text{Screen repair}, \text{Battery replacement}, \text{Full replacement}\}.$$

Step 3: Define event A="at least one claim"=

$$A = \{\text{Screen repair}, \text{Battery replacement}, \text{Full replacement}\}.$$

Step 4: Define event B="catastrophic claim"=

$$B = \{\text{Full replacement}\}.$$

Step 5: Interpret: the same facts produce different events because the insurer, the customer, and the product designer care about different questions.

Risk / actuarial interpretation: If the event is unclear, every later calculation can be mathematically correct and still be useless.

Worked Example

Problem. A hospital backup generator either (i) starts normally, (ii) starts late but works, or (iii) fails completely. Why might an engineer and an insurer define different events even though the sample space is the same?
Micro-steps

Step 1: Sample space:

$$S = \{\text{Normal}, \text{Late start}, \text{Fail}\}.$$

Step 2: Engineer's event: $E = \{\text{Late start}, \text{Fail}\}$ because any operational weakness matters.

Step 3: Insurer's event: $F = \{\text{Fail}\}$ because payout may trigger only under complete failure.

Step 4: Conclude that mathematics does not tell us *which* event to study; the risk question does.

Common Mistake

Do not jump to numbers before defining the sample space. Confusion about outcomes is one of the fastest ways to misuse probability formulas.

4 B. Experimental probability: learning from data

When we use past observations to estimate the chance of an event, we use **experimental probability** (also called relative frequency).

Experimental probability formula

$$\mathbb{P}(A) = \frac{\text{number of times event } A \text{ occurred}}{\text{total number of observations}}.$$

Risk / actuarial interpretation: This is a *data-driven estimate*, not a logical certainty.

Worked Example

Problem. From the opening table, find the experimental probability of *serious disruption* (heavy rain or cancellation).

Micro-steps

Step 1: Relevant outcomes: heavy rain and cancellation.

Step 2: Add their frequencies: $7 + 3 = 10$.

Step 3: Divide by the total number of event dates: 50.

Step 4: Compute:

$$\mathbb{P}(\text{serious disruption}) = \frac{10}{50} = 0.20 = 20\%.$$

Risk / actuarial interpretation: Based on this historical record, about 1 in 5 similar dates may face serious disruption.

Worked Example

Problem. The cancellation outcome happened only 3 times in 50. Why might a risk manager still focus more on cancellation than on light rain, which happened 12 times?

Micro-steps

Step 1: Probability measures *frequency*, not damage size.

Step 2: Compare consequences: light rain costs HK$9,000, cancellation costs HK$92,000.

Step 3: A smaller probability can still create a larger strategic threat if the consequence is much larger.

Step 4: Therefore risk management must eventually combine **likelihood** and **impact**.

Technology Check: Google Gemini + R for relative frequency

Suggested Google Gemini prompt:

`Write R code to compute the probability of serious disruption from the frequencies Dry=28, Light=12, Heavy=7, Cancel=3 and draw a bar chart.`

Example R code:

```
weather <- c(Dry=28, Light=12, Heavy=7, Cancel=3)
p_serious <- (weather["Heavy"] + weather["Cancel"]) / sum(weather)
print(p_serious)
barplot(weather, col=c("skyblue","gold","orange","red"),
        main="Past event outcomes")
```

Investigation / Activity: Why large samples matter

Use technology to compare the stability of an estimated probability when the sample size is 20, 200, and 10,000 simulated observations. Record how much the estimate changes from run to run. What does this suggest about risk models built from short histories? When might a short history still be the best available evidence?

5 C. Theoretical probability and equally likely outcomes

When outcomes are equally likely and can be listed logically, we use **theoretical probability**.

Theoretical probability formula

$$\mathbb{P}(A) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

Risk / actuarial interpretation: This is a *model-based* calculation, not an estimate from data.

Worked Example

Problem. A fair die is used to model project risk levels. Scores 4, 5, and 6 mean serious risk. Find the probability of serious risk.

Micro-steps

Step 1: Sample space:

$$S = \{1, 2, 3, 4, 5, 6\}, \qquad |S| = 6.$$

Step 2: Favourable outcomes:

$$A = \{4, 5, 6\}, \qquad |A| = 3.$$

Step 3: Apply the formula:

$$\mathbb{P}(A) = \frac{3}{6} = \frac{1}{2} = 0.5.$$

Step 4: Interpret: serious risk occurs half the time in this simple model.

Worked Example

Problem. Two factory sites each have a risk level from 1 to 6, with all levels equally likely. If level 1 means *flood-free*, find the probability that both sites are flood-free.

Micro-steps

Step 1: Use ordered pairs (x, y), where x is Site A and y is Site B.

Step 2: Total outcomes: $6 \times 6 = 36$.

Step 3: Only one outcome gives both sites level 1: $(1, 1)$.

Step 4: Therefore,

$$\mathbb{P}(\text{both flood-free}) = \frac{1}{36} \approx 0.0278.$$

Risk / actuarial interpretation: A joint event is often much rarer than each single event by itself.

Thought Trigger

If the two sites are near the same river, should we still treat all 36 ordered pairs as equally likely? A formula can be correct while the model behind it is foolish.

6 D. Event relationships: complement, union, intersection, and Venn thinking

Cognitive Trigger

Let A="rolling an even number" and B="rolling a number greater than 3" on a fair die. Why does simple addition $P(A) + P(B)$ overcount? Which outcomes are being counted twice?

To work confidently with events, use both symbols and pictures. Venn diagrams help you see whether events overlap, stay separate, or cover the whole sample space.

Three basic relationships

- **Complement** A': event A does *not* occur.

$$\mathbb{P}(A') = 1 - \mathbb{P}(A).$$

- **Union** $A \cup B$: at least one of A or B occurs.
- **Intersection** $A \cap B$: both A and B occur.

Addition law of probability

A key rule for combining events is the addition law:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Risk / actuarial interpretation: The subtraction term prevents double counting.

Proof Decision Points ($\star\star$)

Gate 1 (the double-counting trap): To find $P(A \cup B)$, add the two probabilities and then subtract the overlap. If the overlap is ignored, the same outcomes are counted twice.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Only when the events are mutually exclusive does the overlap term become zero.

Self-Check (Detailed Working)

Let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$ on a fair die. Then $P(A) = 3/6$, $P(B) = 3/6$, and $P(A \cap B) = 2/6$. Hence

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \mathbf{\frac{4}{6}}.$$

Direct check: $A \cup B = \{2, 4, 5, 6\}$, which really contains 4 outcomes.

Worked Example

Problem. Suppose the probability of a severe claim in a year is 0.08. Find the probability of *no* severe claim.

Micro-steps

Step 1: Let A= severe claim occurs.

Step 2: Then A'= no severe claim occurs.

Step 3: Use the complement rule:

$$\mathbb{P}(A') = 1 - 0.08 = 0.92.$$

Risk / actuarial interpretation: If severe risk is 8%, the firm still expects a 92% chance of avoiding it in a single year.

Worked Example

Problem. Structural failure has probability 0.012, equipment malfunction has probability 0.045, and both together have probability 0.003. Find the probability of at least one problem.

Micro-steps

Step 1: Start with the addition law:

$$\mathbb{P}(S \cup M) = \mathbb{P}(S) + \mathbb{P}(M) - \mathbb{P}(S \cap M).$$

Step 2: Substitute:

$$0.012 + 0.045 - 0.003.$$

Step 3: Compute:

$$\mathbb{P}(S \cup M) = 0.054.$$

Step 4: Convert to percentage: 5.4%.

Risk / actuarial interpretation: Ignoring overlap would overstate risk and may lead to unfair pricing.

Worked Example

Problem. A student says, "Cyber attack chance is 6%, system outage chance is 5%, so together they threaten us 11% of the time." Explain why this is mathematically dangerous.

Micro-steps

Step 1: The two events may overlap.

Step 2: If some cyber attacks *cause* outages, counting them separately may double count those days.

Step 3: Therefore the probability of "attack or outage" is not automatically $0.06 + 0.05$.

Step 4: The intersection $\mathbb{P}(A \cap B)$ must be considered.

Mutually exclusive vs complementary

Two important event relationships are **mutually exclusive** events and **complementary** events.

- Events A and B are **mutually exclusive** if they cannot happen at the same time. Then $A \cap B = \emptyset$, so
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$
- Events A and A' are **complementary**. They cover all possibilities and cannot happen together.

Investigation / Activity: Use a Venn diagram, not just a formula

Draw a Venn diagram for these three situations:

1. mutually exclusive events,
2. overlapping events,
3. complementary events.

Then write one real-life insurance or finance story for each picture. Which picture is most dangerous to misunderstand when pricing risk?

7 E. Conditional probability, multiplication law, independence, and Bayes' bridge

Cognitive Trigger

A card is drawn from a standard deck. Before any extra information, the probability of drawing a King is 4/52. But if you are told the card is a face card, should the probability still be 4/52? What exactly has changed — the card, or your sample space?

Sometimes one event changes what we believe about another. The phrase **"given that"** changes probability.

Conditional probability

A core rule for conditional probability is

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A),$$

which leads directly to the definition

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}, \qquad \mathbb{P}(A) > 0.$$

Worked Example

Problem. Suppose 22 of the 50 event nights had some rain at all, and among those 22 rainy nights, 5 escalated to cancellation. What is the probability of cancellation *given that* rain has already started?

Micro-steps

Step 1: Let $A=$ cancellation. Let $B=$ rain has started.

Step 2: Use the rainy nights as the reduced sample space.

Step 3: Frequency version:

$$\mathbb{P}(A \mid B) = \frac{5}{22} \approx 0.2273.$$

Step 4: Interpret: once rain has started, the chance of cancellation rises to about 22.7%.

Independence and multiplication law

If two events are independent, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

This identity can also be used as a practical test for independence.

Worked Example

Problem. Unit 1 fails with probability 0.02 and Unit 2 fails with probability 0.05. Assume failures are independent. Find the probability both fail.

Micro-steps

Step 1: Recognise the word *independent*.

Step 2: Apply the multiplication law:

$$\mathbb{P}(\text{both fail}) = 0.02 \times 0.05 = 0.001.$$

Step 3: Convert to percentage: 0.1%.

Risk / actuarial interpretation: A very small joint probability can still matter if the joint loss is huge.

Do not confuse independence with mutual exclusion

It is very important to distinguish **independent events** from **mutually exclusive events**.

- **Mutually exclusive**: events cannot happen together.
- **Independent**: the occurrence of one does not affect the probability of the other.

These are *not* the same idea.

Worked Example

Problem. On a fair die, let A="the score is odd" and B="the score is prime". Are A and B independent?

Micro-steps

Step 1: Odd scores: $A = \{1, 3, 5\}$, so $\mathbb{P}(A) = 3/6 = 1/2$.

Step 2: Prime scores: $B = \{2, 3, 5\}$, so $\mathbb{P}(B) = 3/6 = 1/2$.

Step 3: Intersection: $A \cap B = \{3, 5\}$, so $\mathbb{P}(A \cap B) = 2/6 = 1/3$.

Step 4: Compare:

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \neq \frac{1}{3}.$$

Step 5: Therefore A and B are **not independent**.

Risk / actuarial interpretation: Even very simple events can fail to be independent. Independence should not be guessed by intuition.

Common Mistake

Students should not determine whether two events are independent by intuition alone. Always test independence using probabilities, not just intuition.

Proof Decision Points (⋆⋆)

Gate 2 (independence vs. mutual exclusion): These are not merely different words; they are opposite intuitions.

- If events are **mutually exclusive**, the occurrence of one makes the other impossible.
- If events are **independent**, the occurrence of one leaves the probability of the other unchanged.

So mutually exclusive events with positive probability cannot be independent.

Paired Examples: Works vs Fails

Independent example: rolling a die and flipping a coin.
Mutually exclusive example: rolling a 1 and rolling a 6 on the same single die throw.

Tree diagrams for finite outcome problems

When a problem has several stages, a tree diagram is an efficient way to organise all possible outcomes. It is not a decorative sketch: it is a structured thinking tool.

Worked Example

Problem. A simple fraud screen labels a transaction as *high risk* with probability 0.12. If a transaction is high risk, it enters manual review; otherwise it passes automatically. If a transaction is high risk, the chance it is truly fraudulent is 0.25. If it is not flagged high risk, the chance it is fraudulent is 0.01. Find the probability that a random transaction is fraudulent.
Micro-steps

Step 1: First branch: High risk (0.12) or not high risk (0.88).

Step 2: Second branch: Fraud given high risk (0.25), fraud given not high risk (0.01).

Step 3: Multiply along each branch:

$$\mathbb{P}(\text{High risk and Fraud}) = 0.12 \times 0.25 = 0.03,$$

$$\mathbb{P}(\text{Not high risk and Fraud}) = 0.88 \times 0.01 = 0.0088.$$

Step 4: Add the fraud branches:

$$\mathbb{P}(\text{Fraud}) = 0.03 + 0.0088 = 0.0388.$$

Risk / actuarial interpretation: The tree diagram separates the problem into clear stages and prevents mixing up conditional statements.

Bayes' theorem: a bridge to Module 1

Bayes' theorem is a natural extension of conditional probability, so we include it here as a forward-looking topic that helps you practise updating beliefs when new evidence appears.
In its simplest form:

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Risk / actuarial interpretation: Bayes tells us how to **update a belief** when new evidence arrives.

Worked Example

Problem. Suppose only 2% of devices are truly faulty. A screening test flags a faulty device with probability 0.90, and wrongly flags a healthy device with probability 0.08. If a device is flagged, what is the probability it is actually faulty?

Micro-steps

Step 1: Let F= device faulty, P= flagged positive.

Step 2: Known values:

$$\mathbb{P}(F) = 0.02, \qquad \mathbb{P}(P \mid F) = 0.90, \qquad \mathbb{P}(P \mid F') = 0.08.$$

Step 3: First find the total positive probability:

$$\mathbb{P}(P) = \mathbb{P}(P \mid F)\mathbb{P}(F) + \mathbb{P}(P \mid F')\mathbb{P}(F')$$

$$= 0.90(0.02) + 0.08(0.98) = 0.018 + 0.0784 = 0.0964.$$

Step 4: Now apply Bayes:

$$\mathbb{P}(F \mid P) = \frac{0.90(0.02)}{0.0964} \approx 0.1867.$$

Risk / actuarial interpretation: Even a positive result does not guarantee the device is faulty. Rare events create Bayesian surprises.

Deepening Layer (⋆⋆)

Bayes through counts instead of symbols: Imagine 1,000 devices if the fault rate is 2%. Then about 20 are truly faulty and 980 are healthy. If the test catches 90% of faulty devices, that gives about 18 true positives. If it wrongly flags 8% of healthy devices, that gives about 78.4 false positives. The flagged group is therefore dominated by non-faulty devices. This is the *base-rate effect* in action.

Google Gemini Exploration (Copy & Paste Prompt)

Prompt you can copy and adapt:

```
Generate a Bayes theorem example for gifted secondary students in a risk management context where
a test seems highly accurate but the posterior probability remains surprisingly low.  Explain the
role of the base rate in plain language.
```

Investigation / Activity: Prior probability vs posterior worry

Ask Google Gemini to generate a numerical example where a test is "very accurate" but the posterior probability after a positive test is still below 50%. Then explain why this happens in words. What does this teach us about risk alerts, screening tools, and false confidence?

8 F. Random experiments and random variables

Before introducing random variables, it helps to make the idea of a random experiment explicit.

When is an experiment random?

An experiment can be considered **random** if:

1. it can be performed repeatedly under the same condition(s),
2. all possible outcomes are obtainable and there is more than one possible outcome,
3. the actual outcome is unknown before the experiment is performed.

Discrete random variable

A **discrete random variable** assigns a number to each outcome of a random experiment and takes values in a finite or countably infinite set.

Worked Example

Problem. Let X be the number of heads in two fair coin tosses.
Micro-steps

Step 1: Sample space:

$$S = \{HH, HT, TH, TT\}.$$

Step 2: Apply the rule "count the number of heads":

$$HH \mapsto 2, \quad HT \mapsto 1, \quad TH \mapsto 1, \quad TT \mapsto 0.$$

Step 3: Therefore the possible values of X are

$$X \in \{0, 1, 2\}.$$

Risk / actuarial interpretation: This turns a verbal uncertainty into a quantity we can graph, average, and compare.

Worked Example

Problem. Why is "lifetime of a light bulb" usually treated as a continuous random variable but "number of claims this week" treated as a discrete random variable?
Micro-steps

Step 1: Number of claims is counted in whole numbers.

Step 2: Lifetime can in principle take any real value in an interval.

Step 3: Therefore one is naturally discrete and the other is naturally continuous.

Step 4: In risk management language: claim *frequency* is often discrete, while claim *severity* is often continuous.

9 G. Probability distributions: table, graph, formula, and notation

A discrete probability distribution should be represented in several ways: **tables**, **graphs**, and **mathematical formulae**. You will use all three representations in this lecture.

Notation matters

Remember the notation: capital letters such as X denote random variables, while lower-case letters such as x denote specific possible values of the random variable.

Distribution requirements

A discrete probability distribution for X is valid if:

1. $0 \le \mathbb{P}(X = x_i) \le 1$ for all possible values x_i,
2. $$\sum_i \mathbb{P}(X = x_i) = 1.$$

Worked Example

Problem. A gadget insurer models one-year payout X as follows:

x	0	60	300
$\mathbb{P}(X = x)$	0.82	0.12	0.06

Check that this is a valid distribution and describe how it would look in a bar chart.
Micro-steps

Step 1: Each probability lies between 0 and 1.

Step 2: Add them:
$$0.82 + 0.12 + 0.06 = 1.00.$$

Step 3: Therefore it is valid.

Step 4: In a bar chart, the bar at $x = 0$ is far tallest, the bar at $x = 60$ is smaller, and the bar at $x = 300$ is smallest.

Risk / actuarial interpretation: Tables show numbers clearly, graphs reveal shape quickly, and formulas support calculation efficiently.

Technology Check: R / spreadsheet representation

```
outcomes <- c(0, 60, 300)
probs <- c(0.82, 0.12, 0.06)
barplot(probs, names.arg=outcomes, col="steelblue",
        main="Probability distribution of payout X",
        xlab="Payout", ylab="Probability")
```

Investigation / Activity: One random variable, many representations

Take the random variable “number of late students tomorrow” for your class. Invent a plausible discrete distribution. Then represent it in:

1. a table,
2. a quick bar chart,
3. a sentence in words,
4. symbolic notation, for example $\mathbb{P}(X = 0) = \cdots$.

Which representation made you notice a modelling problem first?

10 H. Expectation, variance, standard deviation, and transformation rules

Cognitive Trigger

You sell 1,000 insurance policies. A claim occurs with probability 1%, and each claim costs HK$50,000. How do you turn a chance mechanism into a money quantity that can guide premium setting, profitability, and capital planning?

Expectation, variance, and the transformation rules for $aX + b$ are worth introducing now because risk management decisions quickly lead to two important questions:

- What is the long-run average outcome?
- How much do outcomes fluctuate around that average?

Link to Risk Management & Actuarial Science

Why actuaries care immediately: Expected value helps with pricing, but variance and standard deviation help with solvency, capital buffers, and the psychological reality that managers must survive short-run bad luck long enough to enjoy long-run averages.

Expected value

For a discrete random variable X,

$$\mathbb{E}(X) = \sum x\mathbb{P}(X = x).$$

Risk / actuarial interpretation: In insurance, this is often interpreted as the **pure premium** or average payout.

Worked Example

Problem. Using the gadget payout distribution $X \in \{0, 60, 300\}$ with probabilities $0.82, 0.12, 0.06$, find the expected payout.

Micro-steps

Step 1: Multiply each outcome by its probability:

$$0 \cdot 0.82, \qquad 60 \cdot 0.12, \qquad 300 \cdot 0.06.$$

Step 2: Compute the products: 0, 7.2, and 18.

Step 3: Add them:

$$\mathbb{E}(X) = 0 + 7.2 + 18 = 25.2.$$

Risk / actuarial interpretation: On average, the insurer expects to pay HK$25.20 per policy.

Second moment, variance, and standard deviation

The Module 1 extract explicitly mentions $\mathbb{E}(X^2)$ and the identity

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2.$$

Equivalently,

$$\mathrm{Var}(X) = \sum (x-\mu)^2 \mathbb{P}(X=x), \qquad \mu = \mathbb{E}(X),$$

with standard deviation

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Worked Example

Problem. Compare two investments with the same expected value but different risk.
Investment A pays +120 or −70 with equal probability.
Investment B pays +40 or +10 with equal probability.

Micro-steps for Investment A

Step 1: Mean:

$$\mathbb{E}(A) = 120(0.5) + (-70)(0.5) = 25.$$

Step 2: Deviations from mean: 95 and −95.

Step 3: Squared deviations: 9025 and 9025.

Step 4: Variance:

$$\mathrm{Var}(A) = 9025(0.5) + 9025(0.5) = 9025.$$

Step 5: Standard deviation:

$$\mathrm{SD}(A) = 95.$$

Quick comparison for Investment B

- $\mathbb{E}(B) = 25$ as well.
- Deviations are 15 and −15, so

$$\mathrm{Var}(B) = 225, \qquad \mathrm{SD}(B) = 15.$$

Risk / actuarial interpretation: Same average does not mean same risk. Risk managers care about spread, not only the mean.

Transformation rules from Module 1

Two powerful transformation rules are

$$\mathbb{E}(aX+b) = a\mathbb{E}(X) + b,$$

$$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X).$$

These are powerful because many financial and insurance quantities are just transformed versions of simpler random variables.

Proof Decision Points (⋆⋆)

Gate 3 (why variance scales with a^2): If every payout is multiplied by a, every deviation from the mean is also multiplied by a. Because variance squares those deviations, the scaling becomes a^2, not just a.

Self-Check (Detailed Working)

Suppose $E(X) = 50$ and $Var(X) = 16$. Let $Y = 3X - 10$. Then

$$E(Y) = 3E(X) - 10 = 3(50) - 10 = \mathbf{140},$$

$$Var(Y) = 3^2 Var(X) = 9 \times 16 = \mathbf{144}.$$

Technology / Computation Box

R code (Expectation and Variance):

```
x <- c(50000, 0)
p <- c(0.01, 0.99)
E_X <- sum(x * p)
E_X2 <- sum(x^2 * p)
Var_X <- E_X2 - E_X^2
cat("Expected Value:", E_X, "\nVariance:", Var_X)
```

Governance check: $E(X)$ helps estimate the fair premium, while $Var(X)$ helps indicate how much uncertainty remains after pricing.

Worked Example

Problem. Let X be the score from one fair die. Define $Y = 2X$. Use the transformation rules to compare $\mathbb{E}(Y)$ and $\text{Var}(Y)$ with $\mathbb{E}(X)$ and $\text{Var}(X)$.

Micro-steps

Step 1: First note that $Y = 2X + 0$, so $a = 2$, $b = 0$.

Step 2: Apply expectation rule:

$$\mathbb{E}(Y) = 2\mathbb{E}(X).$$

Step 3: Apply variance rule:

$$\text{Var}(Y) = 2^2\text{Var}(X) = 4\text{Var}(X).$$

Step 4: Interpretation: doubling every outcome doubles the average but quadruples the variance.

Risk / actuarial interpretation: Leveraging positions often behaves exactly like this: return may scale linearly, but risk scales faster.

Worked Example

Problem. A policy with payout random variable X is modified by adding a fixed handling fee of HK$30 to every claim payment. Let the new payment be $Y = X + 30$. What happens to expectation and variance?

Micro-steps

Step 1: Here $a = 1$, $b = 30$.

Step 2: Therefore,

$$\mathbb{E}(Y) = \mathbb{E}(X) + 30.$$

Step 3: And

$$\text{Var}(Y) = \text{Var}(X).$$

Risk / actuarial interpretation: Adding a constant shifts the average but does not change the spread. This is conceptually important when separating *pricing level* from *riskiness*.

Thought Trigger

A deal that improves expected value may still worsen variance. A deal that keeps variance unchanged may still become expensive. Which one matters more depends on the institution's objective, capital strength, and tolerance for regret.

11 I. Continuous random variables (brief preview)

Some quantities are not naturally counted in whole numbers, such as exact waiting time until an accident, exact rainfall, or exact loss amount. These are **continuous random variables**. For continuous variables, probability belongs to **intervals**, not single points.

Preview idea

For a continuous random variable, $\mathbb{P}(X = x) = 0$ for any exact value x, but interval probabilities such as $\mathbb{P}(1 \leq X \leq 2)$ can be positive.

Investigation / Activity: From histogram to smooth curve

Use R or a spreadsheet to generate 50 waiting times between claims and draw a histogram. Then repeat with 10,000 values and a smaller bin width. What visual change do you observe, and how does that support the idea of a continuous distribution? Why might this matter when modelling rainfall, returns, or waiting time?

12 Student Summary

- Probability provides a structured language for uncertainty.
- Set language, union, intersection, complement, addition law, multiplication law, independence, and conditional probability help you describe complex events clearly.
- Bayes' theorem, discrete random variables, probability distributions, expectation, and variance help you move from event questions to decision-making questions.
- Mutually exclusive events and independent events are very different ideas.
- Venn diagrams and tree diagrams are powerful thinking tools, not childish shortcuts.
- A random variable turns stories into numbers; a probability distribution organises those numbers; expectation and variance turn them into decision support.
- The most important habit is not only calculating correctly, but asking whether the model assumptions deserve to be believed.

13 Review Set 3A (Foundation)

1. In 400 claims, 18 are severe. Find the experimental probability of a severe claim.
2. A fair six-sided die is rolled once. Find the probability of rolling a number greater than 4.
3. If $\mathbb{P}(A) = 0.23$, find $\mathbb{P}(A')$.
4. For events A and B, suppose $\mathbb{P}(A) = 0.35$, $\mathbb{P}(B) = 0.20$, and $\mathbb{P}(A \cap B) = 0.06$. Find $\mathbb{P}(A \cup B)$.
5. Explain in words the difference between **mutually exclusive** and **independent** events.
6. Two independent alarms fail with probabilities 0.03 and 0.08. Find the probability both fail.

7. Let X be the number of heads in three fair coin tosses. List the possible values of X.
8. A payout random variable has values 0 and 120 with probabilities 0.9 and 0.1. Find $\mathbb{E}(X)$.
9. If $Y = 3X$, express $\mathbb{E}(Y)$ and $\text{Var}(Y)$ in terms of $\mathbb{E}(X)$ and $\text{Var}(X)$.
10. In one sentence, explain why standard deviation is useful in risk management.

14 Review Set 3B (Challenge)

1. On a fair die, let A=“odd” and B=“prime”. Show that A and B are not independent.
2. In a class of 120 students, 70 study data science, 46 study finance, and 22 study both. Find the probability that a randomly chosen student studies neither.
3. A disease test is positive in 8% of a population. Among positive cases, 35% are severe. Write the conditional probability statement and interpret it.
4. A random variable has values -40, 20, and 100 with probabilities 0.2, 0.5, and 0.3. Find $\mathbb{E}(X)$.
5. Design two investments with the same expected value but very different standard deviations. Explain why a risk manager may prefer one over the other.
6. Suppose only 1% of transactions are fraudulent. A screening rule catches 90% of frauds but also wrongly flags 5% of legitimate transactions. If a transaction is flagged, is it more likely to be fraudulent or legitimate? Set up the Bayes calculation.
7. Use Google Gemini or R to simulate 10,000 trials of a simple binary event with probability 0.25 and plot the running relative frequency. Describe what the graph shows and what it does *not* show.
8. Write a short paragraph responding to the claim: “A model is useful only if every assumption is realistic.” Do you agree fully, partly, or not at all?

15 Closing Synthesis

Structural Core ($\star$)

Professional storyline (probability in action):

1. Define the sample space carefully.
2. Combine events correctly: use the addition law for *or*, subtraction of overlap included.
3. Update with evidence: use conditional probability and, when appropriate, Bayes’ theorem.
4. Price the uncertainty: convert event-level thinking into random variables, expectation, and variance.

Pause-and-Think

Final challenge for yourself:

- Why must you test whether events are independent before multiplying probabilities?
- Why can a profitable insurance portfolio in expected value still be strategically dangerous if the variance is large?